

The AEC Framework

Advanced Reference — Quantitative Depth

AEC Mentorship Program · Advanced Tier

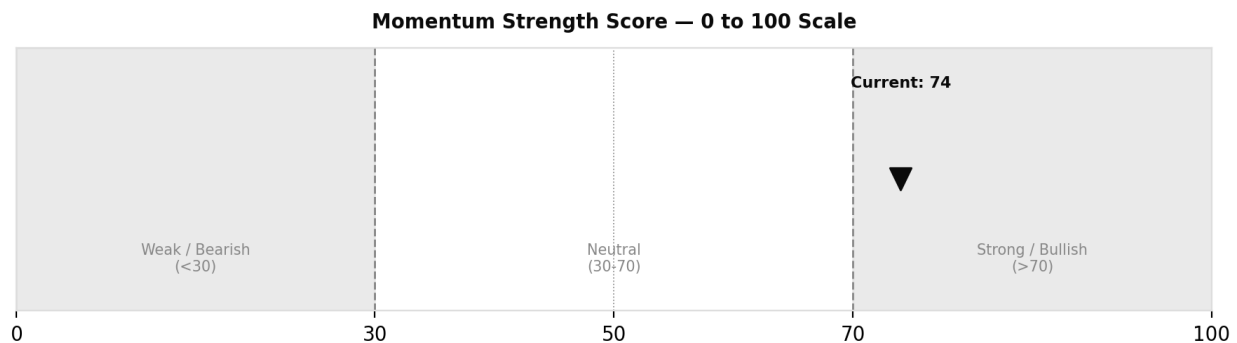
This guide goes beyond the core three-layer walkthrough into the actual quantitative mechanics behind each layer — the real formulas, how they're calculated, how they combine into a single weighted signal, and the position sizing and risk math that determines whether an edge actually survives contact with a real account. This is reference material for traders who already understand the three layers conceptually and are ready for the math underneath them.

I. ADVANCED MOMENTUM SCORING

Rather than a binary “trend up / trend down” read, momentum strength is scored on a continuous 0-100 scale, calculated the same way a Relative Strength Index is built:

$$RS = \text{Average Gain (over N periods)} / \text{Average Loss (over N periods)}$$

$$\text{Momentum Score} = 100 - (100 / (1 + RS))$$



A score above 70 confirms strong bullish momentum; below 30 confirms strong bearish momentum. The 30-70 zone is neutral — Layer I does not generate a directional signal here, regardless of what price action looks like visually.

Multi-Timeframe Weighting

Advanced application weights the higher timeframe score more heavily than the lower timeframe score, since bias should dominate the final read: $\text{Weighted Momentum} = (\text{HTF Score} \times 0.65) + (\text{LTF Score} \times 0.35)$. A trade is only considered when the weighted result exceeds 60 in either direction.

II. ADVANCED STATISTICAL DEVIATION

The core Z-score formula measures deviation in standard deviations from a rolling mean:

$$Z = (\text{Price} - \text{Rolling Mean}) / \text{Rolling Standard Deviation}$$

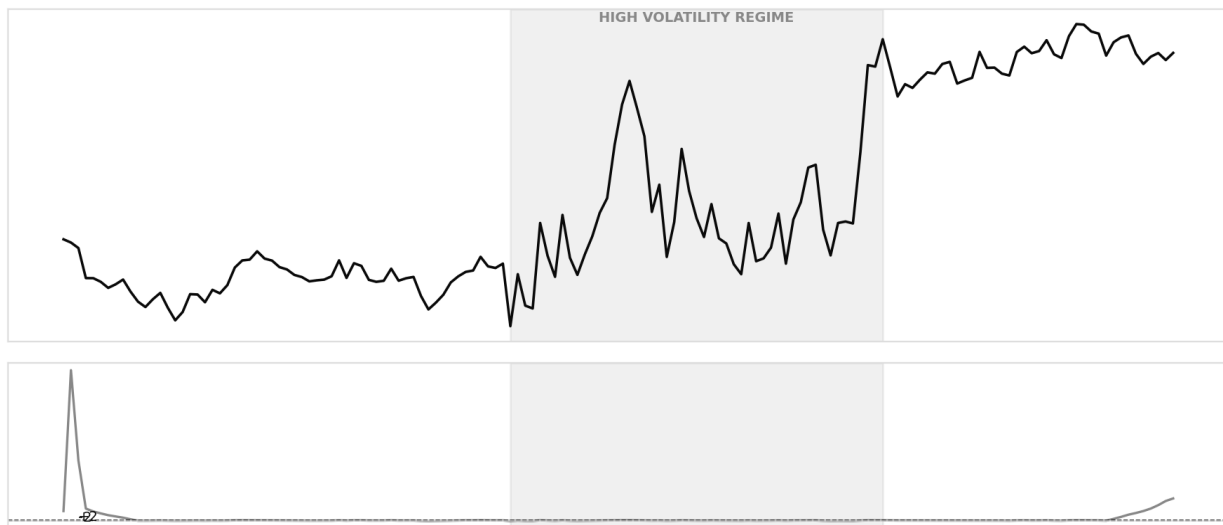
$$\text{Rolling Mean} = \text{Sum}(\text{Price}, N \text{ periods}) / N$$

$$\text{Rolling Std Dev} = \sqrt{(\text{Sum}((\text{Price} - \text{Mean})^2, N \text{ periods}) / N)}$$

Volatility Regime Adjustment

The advanced application of Layer II accounts for the fact that a fixed Z-score threshold behaves very differently in a low-volatility regime versus a high-volatility regime. The same numeric Z-score of 2.0 can represent a minor move in a volatile stretch of the market and a major statistical outlier in a calm one. To correct for this, the rolling standard deviation itself is tracked over time — when the current standard deviation is meaningfully above its own longer-term average, thresholds are widened (e.g., from ± 2.0 to ± 2.5) to avoid over-triggering during genuinely volatile conditions.

Z-Score Behavior Across Volatility Regimes



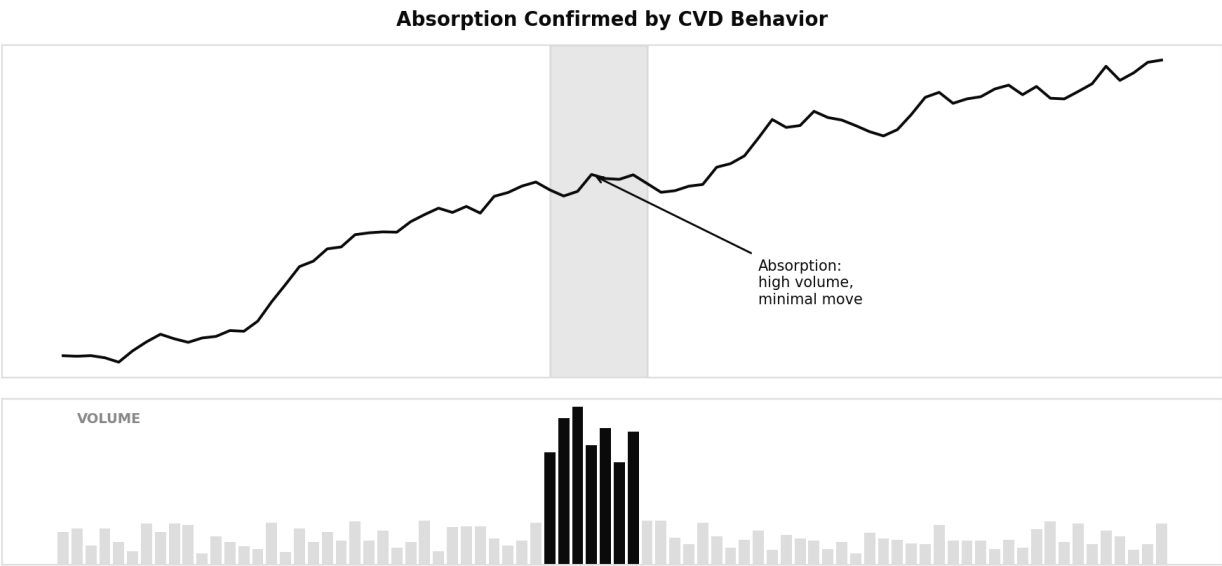
The shaded region shows a higher-volatility stretch. Notice the Z-score swings more violently through this window — without a regime adjustment, this period alone would generate far more (lower-quality) signals than the calmer periods on either side.

III. ADVANCED ORDER FLOW MECHANICS

Cumulative Volume Delta (CVD)

CVD is a running total of aggressive buying volume minus aggressive selling volume across a session. It is the single most important order-flow metric for spotting divergence between what price is doing and what real participants are actually doing underneath it.

Delta (per bar) = Aggressive Buy Volume – Aggressive Sell Volume
CVD = Running Sum of Delta across the session



Absorption identified quantitatively: volume in the shaded window runs roughly 3-4x the surrounding average, while price barely moves — the statistical signature of a large resting order absorbing aggressive flow without breaking.

Institutional Footprint Score

Advanced application combines three order-flow signals into a single composite score (0-100):

Large Orders Present	Up to 45 points — scaled by count and relative size of resting orders vs. book average
CVD Trend Strength	Up to 30 points — scaled by magnitude and consistency of CVD direction
Absorption Confirmed	25 points — binary, awarded only when volume and price-movement criteria are both met

A footprint score of 60 or higher is required before Layer III is considered confirmatory in the aggregation logic below.

IV. POSITION SIZING & RISK MATH

Kelly Criterion (Fractional)

Position size is not arbitrary — it is derived mathematically from your actual edge, using a fractional Kelly approach to avoid the excessive volatility of full Kelly sizing:

$$\text{Kelly \%} = \text{Win Rate} - [(1 - \text{Win Rate}) / (\text{Avg Win} / \text{Avg Loss})]$$

$$\text{Position Size} = \text{Account Balance} \times (\text{Kelly \%} \times 0.5) - \text{half-Kelly, standard practice}$$

Worked example: 78% win rate, average win \$5,428, average loss \$3,000, \$100,000 account.

$\text{Kelly \%} = 0.78 - [(0.22) / (5428/3000)] = 0.78 - 0.122 = 0.658 \rightarrow 65.8\%$

Half-Kelly position size = \$100,000 × 0.329 = \$32,920

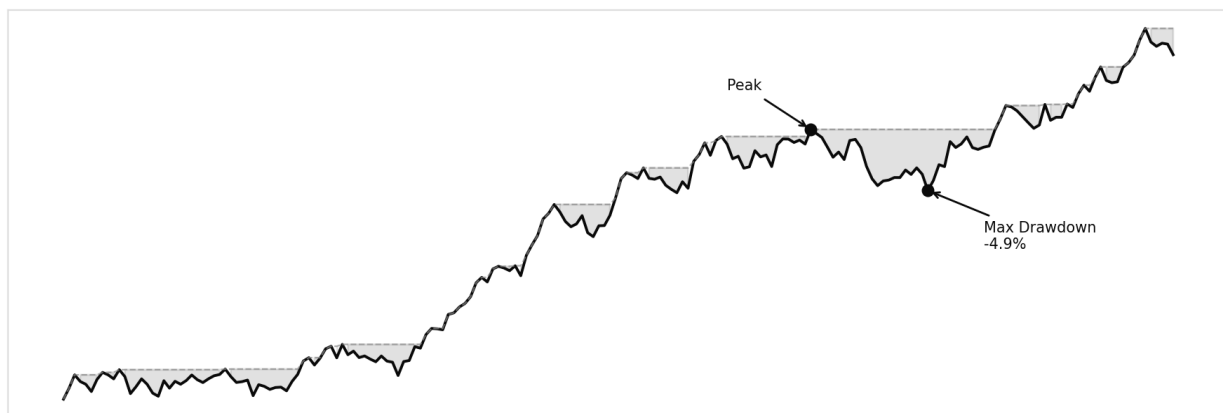
Even at half-Kelly, a 65.8% raw Kelly figure produces a very large recommended position — this is exactly why most professional traders cap position size well below what Kelly technically allows, regardless of how strong the edge appears in the data.

Risk of Ruin

Position sizing decisions should be checked against risk of ruin — the approximate probability of catastrophic account loss given your edge and sizing. A positive edge does not guarantee survival if position sizing is too aggressive relative to the natural variance of the strategy. This is calculated using the ratio of losing to winning probability, adjusted for risk taken per trade, and should trend toward zero as risk-per-trade is reduced — if it isn't near zero at your current sizing, your position size is too large regardless of how good the win rate looks.

V. PERFORMANCE ANALYTICS

Equity Curve — Peak-to-Trough Drawdown



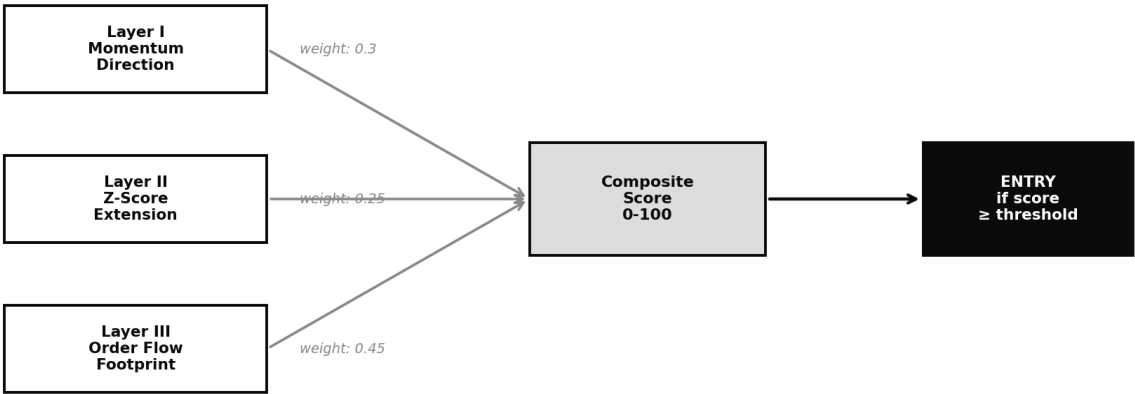
Max drawdown — the largest peak-to-trough decline in account equity — is tracked continuously, not just reviewed after the fact. A strategy with a strong win rate can still carry an unacceptable drawdown profile if position sizing is miscalibrated.

Sharpe Ratio	$\text{Mean}(\text{Excess Returns}) / \text{StdDev}(\text{Excess Returns}) \times \sqrt{(\text{periods per year})}$ — risk-adjusted return
Expectancy	Sum of all trade P&L / Number of trades — the single number that determines long-run viability

Win Rate	Winning trades / Total trades × 100 — meaningful only alongside expectancy, never alone
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VI. FULL SIGNAL AGGREGATION LOGIC

This is how the three layers actually combine into one weighted decision, rather than a simple unanimous-vote requirement:



Layer I Weight	0.30 — directional confirmation, necessary but not sufficient alone
Layer II Weight	0.25 — acts primarily as a filter/veto rather than a positive contributor
Layer III Weight	0.45 — weighted highest, since it provides the most precise, real-time confirmation
Entry Threshold	Composite score must reach 70+ before an entry is generated
Hard Veto	Regardless of composite score, Layer II failing the extension filter blocks entry entirely

A NOTE ON RISK

None of the formulas in this guide guarantee an outcome. They are tools for making consistent, repeatable decisions instead of emotional ones — the math replaces guessing, it does not replace risk. All trading involves substantial risk of loss, including total loss of account balance. Every position sizing formula in this document assumes historical win rate and average win/loss figures remain representative of future performance, which is never guaranteed. Charts shown are illustrative examples, not live market data.

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Ascenda Enterprise Capital LLC · izaiah@ascendaenterprisecapital.com